

Basing off of COMET, Praxis and the ECON526 Github Repo with slides and .html.

This would be a continuation of ECON 326 something like: “you already learned IV and DiD + other techniques like that in COMET, here's how we can use ML to supercharge them and make them more powerful.”

Alternatives currently:

- CPSC 340: Far too technical, more math/CS theory not application and gated behind CS Data Structure and algorithm and Math 200/221.
- ECON 526: Master's course.

Proposal:

- **Notebook 1:** Prediction, Inference and Causality. The ML workflow: train/test split, loss, overfitting, the bias–variance tradeoff, cross-validation. Something simple like KNN (k=1 overfit and k=n underfit) DSCI 100 style. More of a review and warmup
 - Goal apart from warmup: students can run a supervised ML pipeline and explain what overfitting/underfitting is.
- **Notebook 2:** Regularization, Ridge Regression and Lasso. The goal is to handle the "too many controls" problem (OLS breaks when covariates > observations) and introduce regularization as a tool. We use an econ dataset with a lot of covariates to demonstrate this.
 - Goal: student can fit ridge/lasso, read the coefficient path, and articulate "lasso = variable selection."
- **Notebook 3:** Trees, Forests & Boosting. Flexible nonparametric prediction. We start with a single decision tree (and discuss overfitting). We can cover bagging, random forest and some boosting. Make sure to say this is not causal. We do not touch neural nets (can point to the other praxis notebook). We will do a single tree on a small dataset (visualize the splits), then RF on the same data, then test-error comparison.
 - Goal: student can fit a random forest and knows it's a flexible, nonparametric predictor.
- **Notebook 4:** From Prediction to Treatment Effects. More conceptual and very important. We have now learned 3 notebooks of tools, how do they connect to causal questions. If you throw the tools at questions you get bias, how to get at causality. The first 3

notebooks look at predictive power, but that's not causality. We want "what is the effect of D on Y" not "how well can I predict Y"

- We will do this with: Potential outcomes, ATE, unconfoundedness (connect explicitly back to COMET: IV and DiD are the alternative identification strategies for when this assumption fails), regularization and overfitting bias. Small teaser for a fix: orthogonalization and cross-fitting.
- Goal: student can explain the difference between a prediction problem and an identification problem, state the unconfoundedness assumption, and say why bias treatment effects occur if we just run normal ML.
- **Notebook 5:** Post-Double-Selection Lasso. Show first how regular Lasso fails, short notebook on Econ dataset, one clean applied example with Post-Double-Selection Lasso. Side by side comparison showing normal Lasso and Post-Double-Selection Lasso tied back to notebook 4.
 - Goal: student can run Post-Double-Selection Lasso and explain why selecting controls "naively" is biased.
- **Notebook 6:** Modeling Treatment Assignment: Propensity scores / IPW / doubly-robust estimation. Shorter, mostly a set-up for 7 and 8. We will estimate a propensity score computer IPW on the same dataset.
 - Goal: student can estimate a propensity score, compute IPW and AIPW, and explain "doubly robust."
- **Notebook 7:** DML: Estimate something in Econ using any learner. Partially linear model, Frisch–Waugh–Lovell / "residual-on-residual" intuition, the orthogonal moment, cross-fitting. Big notebook of a powerful tool for causal ML
 - Goal: student can set up a DML estimation with any learner, explain orthogonalization and cross-fitting
- **Notebook 8:** Heterogeneous Effects & Causal Forests: shift from "the average effect" to "who benefits." Start with the more intuitive meta-learners, conditional average treatment effect, (policy targeting), causal forest as the generalization of Notebook 3's random forest. Visualize the effect heterogeneity.
 - Goal: students can estimate CATEs with meta-learners and a causal forest, visualize heterogeneity, and articulate what a CATE is and why it's different from an ATE.

Sources/Useful things to base it on:

- <https://www.pywhy.org/EconML/>
- <https://docs.doubleml.org/stable/index.html>
- https://datascience.quantecon.org/applications/ml_in_economics.html
- https://causalml-book.org/chapters/CausalML_book_2022.pdf -> Really good
- <https://matheusfacure.github.io/python-causality-handbook/landing-page.html>
- <https://arxiv.org/pdf/1903.10075> (Machine Learning Methods Economists Should Know About)
- <https://ubcecon.github.io/ECON526/>